“I can’t buy a house! My whole life is ruined!” – Studying impact of Drastic Increase in House Prices for Australian Youth

Evellyn Verity Mesak

Part 1: Introduction

1.a. What happened in Australia in 2020-2024

Australia has faced housing affordability problem since early 2000s, a situation exacerbated by the housing boom of 2020-2024. Traditionally, house prices are affected by changes in population, interest rate dynamics, unemployment, economic activity, housing supply, stock prices, and income levels. When COVID-19 happened, there were expectations that the economic crisis will reduce house price level in Australia.

However, during 2020-2024, prices rose contrary to theoretical expectations. The Australian CPI housing rose from 121.9 to 149 from December 2019 to December 2024, increase 27 points within five years. In the past, the same amount of growth happened in 10 years. Wong (2024) argued that low interest rate and quantitative easing implemented during the pandemic played a significant role in the housing market’s performance. They also argued that unconventional monetary policy such as very low rates and generous fiscal stimulus in 2020-2021 plays significant roles in rising house prices in Australia.

This price surge sharply contrasted with income growth. Using GDP per capita as proxy to measure income growth of Australian, during the same period of 2019-2024 the index only grew by four points. This sharpens the housing affordability problem. In October 2025, Australian Institute of Health and Welfare claim that: “In 2024–25, an estimated 1.26 million low-income households were in financial housing stress, spending more than 30% of their disposable income on housing (ANU CSPR, 2025). One in five (20.5%) households in the rental market were low-income households in financial stress, compared with homeowners with a mortgage (14.7%) and homeowners without a mortgage (0.3%).”

1.b. Access to housing: This is a Lifetime Problem.

ANUpoll of Attitudes to Housing Affordability held in March 2017 shows that 74.7% of Australian perceives that home ownership is part of the Australian part of life. Furthermore, there are many social and economic benefits tied to owning a house. We can group these benefits into three main groups:

1. Security of housing tenure
    a. Stability: Owners have the security of a place to live for as long as they choose, without fear of eviction or unexpected selling.
    b. Control: Unlike renters, owners can make changes and alterations to their homes without needing landlord permission.
2. Social benefits
    a. Community: Owning a home often leads to a stronger sense of community and the ability to build long-term friendships with neighbors.
    b. Settled feeling: Homeowners report feeling more settled and less transient compared to renters.
3. Economic benefits
    a. Wealth building: Homeownership can be a form of wealth building, especially if property values increase over time.
    b. Financial stability: It eliminates the risk of unexpected rent increases or a landlord deciding to sell, providing more predictability in household expenses.
    c. Flexibility: Owning can provide more flexibility for families and individuals who wish to stay in the same area for a long period

1.c. Research Gap and Research Questions

While rising in house unaffordability caused by unconventional monetary policy is discussed in much research, the quantitative model of How COVID price surge affects **lifetime wealth trajectories** for different birth cohorts is still under-discussed. In this essay, I built a computation model that can simulate wealth trajectory of young Australian at retirement compared to their parents.

This model will be used to answer some research question as follow:

Main Research Question:
Assuming retirement age of 67, what is the magnitude of impact of house unaffordability on wealth at retirement of younger generation and their parents (adjusted for inflation)?

Subsequent Research Question:

- How did COVID-19 housing boom worsen the housing affordability problem for young Australian?
- If people buy house at much older age or cannot buy house at all, can current private saving and superannuation scheme afford to cover all expenses during retirement?
- If fewer people can buy house, what are the economic and social implication both in productive years and retirement?

Part 2: Analysis Plan

2.a. Selected Evaluation Method and Comparison to Other Methods

The research questions require projecting wealth outcomes over a 40+ year horizon and comparing entire lifetime trajectories between generations. This necessitates a method capable of dynamic, long-term forecasting.

Method	Best For	Main Advantage	Main Limitation
Cross Tabulation	Describing current patterns	• Simple, • transparent, • no assumption	• Cannot be used for prediction • No causality • Hard to model interaction between interacting variables
Multivariate Regression	Identifying correlations	• Statistical rigour • Control (can isolate one variable or see multiple variables works together) • Less assumption	• Hard to model causality (correlation is not causation) • Can only model limited dynamic
Simulation	Understanding mechanism	• Can estimate future projection • Can model complicated dynamics	• Hard to validate • Complex, might be hard to understand • Many assumption

Therefore, a computational simulation is selected as it is the only method that can answer the "what is the magnitude of impact" on future retirement wealth, which is inherently a forward- and long-term-looking question.

2.b. Example and Counter Example

To study the housing affordability problem, we compare a younger generational cohort to their parents' cohort. To analyse the impact of the housing boom, we then compare younger cohorts who entered the housing market before and after the boom.

2c. Measure of Housing Affordability

To measure housing affordability for individual, I use comprehensive metrics used in various study in housing market such as:

1. Price to Income Ratio: House price / annual income
2. Mortgage Stress: % income spent on mortgage
3. Years to save deposit: time needed to accumulate deposit
4. Qualifying Income: qualifying real household income needed for purchase

2.d. Modelling of Wealth Trajectory

In modelling the wealth trajectory, I calculate elements of wealth for each calendar year in a cohort's life. The wealth elements included in the model are:

- Income: Distribution follows the distribution of income from latest ABS data from 2019-2020. This number then projected using GDP per capita for past years and assumed to have 3% growth yearly in the future.
- Rent: Assumed to be equivalent to $30,000 in 2024 dollar. This number then projected using housing CPI for past years and assumed to have same growth as house price in the future.
- Consumption: minimum consumption assumed to be equivalent to $25,000 per year or 35% of disposable income (income – rent / mortgage) – whichever higher. This number then projected to the past using CPI and assumed to grow alongside total inflation in the future
- Saving: Income – Consumption – Rent/Mortgage, assumed to yield 4.5% per year in return.
- Superannuation: 11.5% of yearly income and produced 4% return yearly.
- House Price: Assumed to be equivalent of $1,002,900 in 2024 value – according to latest housing median from ABS. This number then projected to the past using housing CPI and assumed to have 4% growth in the future.
- Buy house: For each year in simulation, I checked if saving is more than deposit needed for house (10% of price) and yearly mortgage is less than income – consumption. If these two conditions fulfilled, then a cohort can buy a house.
- Mortgage payment, Mortgage Balance: assumed mortgage rate is 6% and average tenure is 30 years.
- Home equity: house price – mortgage balance
- Net worth: saving + superannuation + home equity
- During retirement, consumption is covered using superannuation and savings, if consumption is larger than available fund, then a cohort is assumed to receive age pension to cover the rest.

What works well using this model:

- This model can captured generational wealth gap caused by house affordability problem.
- This model can capture impact on COVID housing boom on the cohorts' net worth.
- This model can capture impact of not having house during retirement – showed that the cohort that are locked out from housing market need extra income from age pension

2.e. Data, Assumption, and Limitation

Data

Data used to create microsimulation model are:

Group	Indicator	Source
Economic Data	CPI	ABS
	Housing CPI	ABS
	GDP per Capita	datacatalog.worldbank.org
Income and wealth distribution	Cross tabulation data on income and wealth	ABS (2005,2015,2020)
Other Parameter	Australian median house price 2024	ABS

Assumption

Some of the assumption not mentioned in part 2.d includes:

- Net worth calculation excludes some asset class that is included in ABS data that is assumed to not produce return (e.g: content dwelling and vehicle)
- Net worth calculation excludes asset class that has high standard error in ABS data such as trusts, incorporated/unincorporated business.
- For cohort simulation, life begin at 25 and assumed to end at 84. Furthermore, at the age 25, average Australian is assumed to have savings equivalent to $35,000 and superannuation equivalent to $17,500 in 2024 dollar, this number is taken from income and wealth data from ABS.
- No cap on age pension, entire income shortfall during pension age will be covered by age pension. This way, age-pension in simulation can also be used to estimate income shortfall caused by housing affordability problem.

Limitation

Limitation on this model includes:

- Assumption heavy, growth projected to be constant and linear.
- Does not consider economic turbulency such as economic crisis or economic boom.
- Individual Heterogeneity Ignored. Ignore difference in behaviour such as saving and consumption habit. Ignore major life events such as marriage, divorce, kids, and inheritance, which has strong correlation to wealth building.
- No behavioural or strategic adaptation. Including changes in house prices due to change in demand, dual income house buying, slower population growth, shock from emigration/immigration.
- Cannot model effect on aggregate level. The effect on overall economy is unknown as this model can only performed on individual.
- Due to limitation of data, I am forced to use central measure (mean and median) interchangeably during validation. However, as wealth and income data are typically right skewed, the mean is sensitive to extreme values and may not represent the typical experience, potentially introducing bias where the median would be more appropriate.

2.f. Model Validation

I use the known wealth trajectory of older 'average Australian' as validation data. To test the soundness of the model, I compare the result produced by model of someone born in 1965 that have median income with the average wealth produced by the ABS.

		Property	Superannuation	Savings	Networth (a+b+c)	Income
2005	Model	124,364	115,983	63,129	303,476	51,565
	Actual from ABS				269,098	51,220
	Error				**-11.3%**	**-0.7%**
2015	Model	469,095	254,950	141,790	865,836	69,298
	Actual from ABS	575,700	213,200	112,900	901,800	91,416
	Error	**23%**	**-16%**	**-20.4%**	**4.2%**	**31.9%**
2020	Model	813,089	357,285	141,790	1,312,164	80,336
	Actual from ABS	884,000	375,300	133,200	1,392,500	82,836
	Error	**8.72%**	**5.04%**	**-6.1%**	**6.1%**	**3.1%**

Table 1: Model validation – Comparing wealth trajectory obtained from model to actual wealth trajectory of older generation

Model Performance Evaluation:

- **Strengths:**
  - High accuracy for individuals closes to retirement age.
  - No systematic bias: The core simulation logic is valid and shows no consistent over- or under-estimation.
- **Key Limitation:**
  - Lacks fidelity in simulating financial behaviour during mid-life
- **Conclusion:** The model is fit-for-purpose in projecting the key metric of **wealth at retirement.**

Part 3: Findings

3.a. Simulation Result

Comparison of Wealth: Snapshot at Retirement

Cohort	Net Worth	House Equity	Super-annuation	Saving	Age Pension *
Parent (Born 1965) - median income	2,205,409	1,209,848	658,965	336,595	0
Young - Born 1985 - median income	1,591,076	0	966,150	624,926	190,973
Young - Born 1985 - 70 pct	4,112,534	2,294,563	1,246,339	571,633	0
Young - Born 1994 - median income	1,630,201	0	1,069,598	560,603	180,411
Young - Born 1994 - 80 pct	5,325,844	2,832,315	1,559,552	933,976	0

***Additional income from age-pension needed to support mortgage/rent+consumption during retirement**

Table 2: Wealth snapshot at retirement – actual dollar

Cohort	Net Worth	House Equity	Super-annuation	Saving	Age Pension*
Parent (Born 1965) - median income	2,019,022	1,283,363	540,844	435,466	0
Young - Born 1985 - median income	1,032,446	0	483,923	548,523	95,654
Young - Born 1985 - 70 pct	2,265,256	2,317,897	624,263	491,697	0
Young - Born 1994 - median income	840,488	0	428,980	411,508	72,357
Young - Born 1994 - 80 pct	2,384,692	3,024,330	625,485	558,458	0

***Additional income from age-pension needed to support mortgage/rent+consumption during retirement**

Table 3: Wealth Snapshot – adjusted to 2024-dollar equivalent

Comparison of Wealth: Lifetime Wealth Trajectory

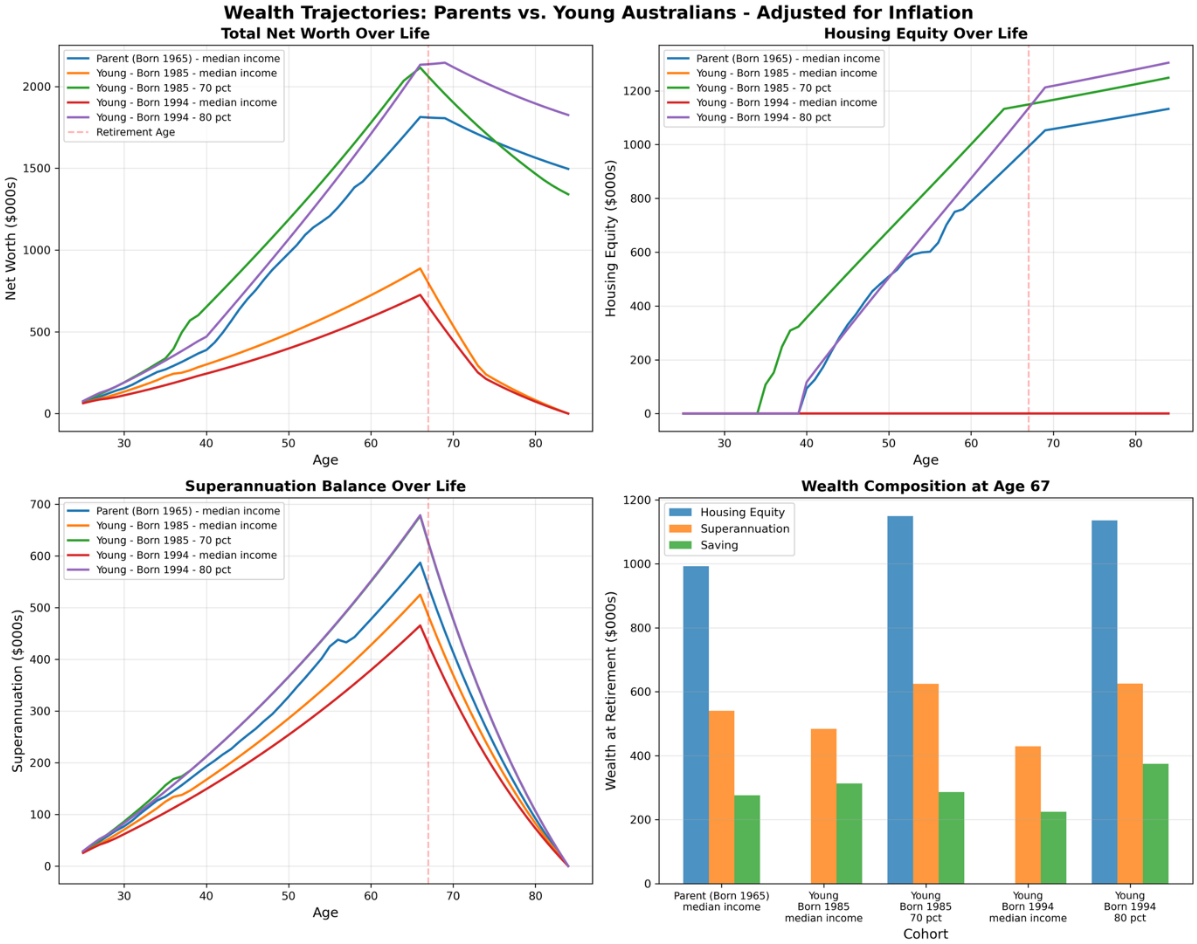

Figure 1: Wealth Trajectory Across Cohort

Housing Affordability Measure Across Cohort (Calculated at first year of home ownership)

	Parent	Young Generation Pre-Covid	Young Generation Post-Covid	Trend
1. Price-to-Income Ratio	975%	959%	1094%	
2. Mortgage Stress	59%	63%	67%	
3. Years to Save Deposit	14	10	14	
4. Qualifying Income	50 Percentile	70 Percentile	80 Percentile	

Table 4: Housing Affordability Measure

3.b. Findings

Different Economic Playing Field

The simulation result has shown that in terms of opportunity to own a house, young Australian and older Australian played in different economic playing fields, shown by these findings:

- **Rising in qualifying income for younger generation: smaller proportion of young generation can afford to buy house through income alone.**
  For younger generation, to be able to buy house in the median price, they need to be making significantly more than the median income. The same opportunity can be afforded by the previous generation by making median income. Furthermore, looking at the housing affordability measure, even if the young generation can enter the housing market, they are saddled with heavier burden of housing cost even with higher income in general.

- **Covid-19 housing boom worsen the Australian housing crisis, creating cohort impact among younger generation, separating those who can buy house before COVID-19 happen and those who cannot.**
  Within the same generation, the hurdle to buy house rise significantly after the COVID-19 housing boom. The older millennial qualifies to buy median house price with 70 percentile income while the younger millennial needs 80 percentile income to afford the same opportunities.

- **Parent generation with median income have more wealth compared to young generation from median income**
  Looking at the total net worth at the time of retirement, the younger generation making median income accumulate significantly less compared to their parents as they miss opportunity of building wealth through home ownership. Furthermore, young Australian produced median income is projected to need additional income from age-pension because they will not own a house by the time they retire, which does not happen for parents' generation with median income.

Part 4: Policy Implication

4.a. Wealth inequality: Larger gap and more concentrated wealth

The findings from micro-simulation shows that home ownership is hard to obtain through income alone, as to be able to afford a median house price, the younger generation must earn significantly above the median income. Furthermore, even if they can buy house, mortgage stress is higher compared to the previous generation due to house price rises faster than wage.

When only small portion within population can afford to buy house purely through income, The "Bank of Mum and Dad" becomes a gatekeeper. Access to homeownership becomes increasingly determined by family wealth, not individual merit or income. This creates an aristocracy of inheritance, cementing inequality and reducing social mobility.

Furthermore, while mortgage payment is ultimately building equity (ownership) in an asset, rent payment is a permanent, non-recoverable cost. It provides a service (shelter) but builds zero equity. It often directly funds the landlord's mortgage, effectively transferring wealth from the asset-poor renter to the asset-rich landlord. This phenomenon causes even larger wealth gap between homeowner and renter.

Lastly, home ownership provides a lot of economic opportunity that is not available to the non-house owner. As home ownership is a vehicle to build wealth, when a generation does not have sufficient access to housing market, less wealth can be inherited to the next generation. Furthermore, Australia has many financial benefits that favours homeowner such as tax deduction for property investment, home equity use, and government support addressed to first home buyer. These in-kind benefits will not be enjoyed by those locked out of housing market.

4.b. Fiscal Time – Bomb on Retired Population

Currently, the Australian Retirement system is built on three pillars: voluntary saving, superannuation, and government pension age. Furthermore, the policy on superannuation is calculated based on assumption of low housing cost in retirement.

The result in simulation shows that for the permanent renter class in young generation, income generated through personal saving and super-annuation is not enough to cover the consumption and rent during retirement years, hence need additional support from age-pension. What is more alarming is based on increase in qualifying income for home ownership, more than 50% of Australian will be a permanent renter, hence more people will be dependent on the pension age.

Furthermore, if Australian have less disposable income due to rent, house affordability problem will degrade Australian living standard during retirement. A study done by Australian treasury shows that even with additional income from age-pension, many pensioners who rented privately had high costs and poor outcomes. Old-age renter reported to experience financial stress and income poverty.

4.c. House Affordability Problem Fragilize Everyone

As shown in the simulation result, with increase magnitude in house affordability problem, there can be two different pathways: become a permanent renter or become a house owner but has significant mortgage burden. Both of this pathway have one thing in common: people can take less risk.

Burdened by high housing cost, young people are less likely to start businesses, take career risks, or pursue innovative but uncertain paths. This makes the overall economy less dynamic and competitive.

This tendency of prioritize job security above everything else will unintentionally put earning cap for the young generation, adding more downward pressure to the house affordability problem.

4.d. Decline of well-being tied to owning a house

Housing is distinct from other asset classes, as it provides both utility and financial benefit. Furthermore, unlike owning stocks, home ownership involves partaking in a local community. On an individual level, it provides social attachment, stability, and security. On a community level, it creates social cohesion; homeowners have longer tenures than renters, which reduces residential churn and facilitates the process of community building.

Part 5: Future Research

As discussed in limitation of this model, there are three opportunities to improve this research with more data:

5.1. Adjust driver assumption on individual heterogeneity, including life events that contribute to wealth building.

While this model is adequate to answer the research questions, the assumption used in building the model might be too simplistic. One of the key weaknesses of this model is that cohort's wealth trajectory is assumed to be an individual, buying house with only one income. This is far from reality, as many people buy house together as a couple. Furthermore, saving and consumption in this model is also modelled increase linearly with income which might be realistic for an individual. However, with family building, consumption pattern will change with marriage/kids. This underlines the importance of fine tuning the model key driver and assumption on individual heterogeneity and major life events.

5.2. Model strategic adaptation

The other area for improvement before we can use this model to study the aggregate effect of change in housing affordability on Australian economy and social dynamic is that we must add some strategic adaptation elements in the model. For example, if less people can afford to buy a house, will the price still increase, or will it reach stagnation? Or will the house affordability problem affect population growth? And how would this dynamic affect the house price if population stagnate? Does immigration/emigration have effect on house price? These kinds of question would be adding needed nuance to understand the aggregated effect of housing affordability problem.

5.3. Study aggregate effect

One of the implications claimed in this research is that housing affordability problem that is worsened by COVID-19 housing boom is retiree who rents will put more burden to taxpayer, as more people will need age pension to help them with rent + consumption during retirement years. The more meaningful question is by how much. Due to limited data on individual, the cohort simulation can only be done assuming the 'average Australian' through cross-tabulation data on wealth and income from ABS. But given more data for different individual, we can observe more trajectory. Given enough sample, we can estimate the aggregate impact of the additional amount needed to support the renter retirees impacted by the house affordability problem.

Furthermore, with some suitable adjustments, this model provides a platform for specific and valuable future policy evaluations. With suitable modifications, it can be used to simulate the impact of housing interventions (such as the First Home Guarantee, First Homeowner Grant, and Stamp Duty concessions) to answer critical research questions like:

- 'What is the comparative effectiveness of different first-home buyer subsidies in improving lifetime wealth outcomes for low-to-median income earners?'
- 'Does lowering the deposit barrier through a guaranteed scheme lead to higher long-term mortgage stress and financial fragility?'
- 'Which policy combination most effectively reduces reliance on the Age Pension by enabling sustainable home ownership?'

Simulating these scenarios would allow policymakers to choose interventions with maximum effectiveness and avoid unintended consequences.

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